
BFMC 2025 : Status Report 5**Date :** Monday, May 19, 2025**Prepared by :** Pothole Avoidance**Reporting Period :** Tuesday, April 22, 2025 - Monday, May 19, 2025

Overview of project status

As was mentioned in the end of the previous status report, the month of May had a heavy focus on integration testing. A detailed plan of the milestones that needed to be reached for seven weeks ahead of the competition was established, with a progressive increase in the scope of the testing and a progressive decrease in the scope of improvements being worked on. The encoder task has been completed and successful results have been obtained. The power board remains an important challenge to this day. On the software level, development has reached a stage where the focus is mainly on bug-fixing and testing for a variety of different situations.

Current Phase

The current phase of the project is a heavy focus on working for reliability. We know what our car is capable of accomplishing, the trouble being its inconsistency in achieving these results. We don't want to arrive at the competition and to have our fingers crossed hoping that « today the car will perform well ». This requires making sure that we have backups for all hardware components, that the software is free of small little bugs and that no major modifications are made after a set deadline.

The plan for the work leading up to the competition was laid out in clear. Every work session had and has a clear objective from the beginning of May to mid June. The objectives for every work session can be found in the exports of the planning¹. Having this planning gave us a clear path for what to accomplish so that we can be confident when we take the plane to Romania.

Task Status - Overview

The past few weeks have been very intense in terms of work for the whole team. Simon, Xin Ya and Yu Qing spent extensive amounts of time working in the lab on all sorts of tasks once school was finished. The priorities were established on the basis of the critical dependencies of our application. These critical dependencies were outlined in the previous status report and concerned odometry. Testing, calibration and analysis of the results of these tests and calibration consumed a great deal of time. These were all required such that the right decisions could be made concerning the choice of power board to use and the choice of which instrumentation feedback to use for our odometry calculation.

Encoder Addition : This task is of high priority and is complete. As was mentioned in the previous status report, preliminary results from the encoder had already been obtained. However, further testing proved that the initial implementation had important shortcomings and revealed other issues in the overall design of the RTOS implementation on the STM32 embedded platform. Modifications to the embedded platform for a real implementation of RTOS made it possible to obtain accurate and real-time measurements of both wheel speed and displacement of the vehicle².

Power Distribution Board Design : The design of the power board has proven to be the most complex task for us this year. Multiple different attempts have been made to design and build a power board that meets our requirements and constraints. Even after four months of work on this task, we still do not have a power board that adequately responds to all our needs. The *Power Board V2*³ was ordered in early May and was received a week later. Preliminary testing proved promising with voltage regulation

¹See *task_tracker_status_5.pdf* and *may_schedule_status_5.pdf*

²See *powerboardV2_testing_report.pdf*

³See *pcb_2025_v2.step*

working well across a wide range of voltages⁴. However, we came to realize that the design was severely limited in its ability to perform under loads. Workarounds using *Power Board V1* were evaluated⁵ and are still in progress.

Hardware Mounts : The intensity with which the testing is ramping up in the weeks ahead of the competition calls for a finalization of hardware mounts. For our testing to be reliable, we need every component to be in a fixed place and to limit any variations in the modifications. As such, mounts for *Power Board V1* were finalized, the adjustable camera mount was replaced for a fixed position mount which is more sturdy and limits vibrations, and the IMU mount was refined. Integrated cable management using vinyl tubing was implemented to limit the number of exposed wires and the possibilities of them being hooked and unplugged accidentally.

Blocking Points and Encountered Issues

RTOS implementation on the STM32 : As we were testing the encoder for different speeds at which the motor was spinning, we ran into issues when attempting to compute the results for higher rotation speeds. Initial attempts were made at reducing the computation time of other tasks by removing heavier functions and decreasing the period size, but we quickly realized that the problem at hand was more fundamental. The implementation of RTOS that came with the embedded platform was a *fake RTOS*. It did not use interrupts to respect deadlines and was basically just sequential execution of different functions. This meant that if a certain function took longer to execute, it would block all other functions from running. This made it impossible for the encoder task to run at its required rate (~100Hz) because some tasks took longer than 10ms to execute. The entire RTOS framework on the embedded platform was therefore redesigned for proper implementation using interrupts and to prevent tasks from getting blocked.

Power Distribution Complexities : As mentioned in the previous status report, one important task was working on a power board redesign with voltage regulation in mind. The issue of current capacity was a central concern in our design, and was made even more complex because of the ambiguous response of the motor manufacturer. We went ahead with the design of the power board and ordered it. However, upon testing the power board under load, we were quick to realize that there was an issue in terms of the current that it could deliver. When under full load (*i.e.* driving the car with all its components mounted), the current supply to the motor was insufficient and the motor stalled. What is more, a mistake in our design also made it such that the 5V voltage regulation was also unable to drive the servo for lack of current. We believe that the cause of these issues is the traces of the power board being too small. This mistake was made because of last minute changes made in the middle of the night right before ordering the power board. Attempts to bypass these traces are in progress, and the use of another less demanding motor is also being considered.

Next Steps

With one month left before the end of the competition, the design approach is changing from a *in depth solution* mindset to a *quick fix* mindset starting next week. The scope of modifications is being greatly reduced and any encountered problems will have to be subject to a decision rapidly. The team must be confident that the in depth work that was done over the past 8 months was of high quality and must refrain from any major design changes. Two final integration tests are planned for the end of May and the beginning of June. These tests will serve as proofs of concepts and fine tuning opportunities right before leaving for Europe. The first two weeks of June will be taken off so that everyone can have a break and change their spirits after having worked very hard in the past two months. This will enable the team to be well rested and focused when the competition arrives in late June.

⁴See *powerboardV2_testing_report.pdf*

⁵See *PB_remake_options.pdf*